

UCI Solar Airplane

Mission Statement

The project seeks to develop a solar airplane that proves solar power can be viable as an energy source, as well as capable of providing assistance in humanitarian aid efforts.

Objectives:

1. Must be functional
2. Must be powered by solar panels and batteries
3. Must have a flight time extended by 10% beyond its battery life in sunny weather conditions
4. Avionics and Propulsion equipment must be integrated and easily accessible throughout the aircraft
5. Must include functional landing gear and be capable of performing takeoff and landing
7. Modular design (standard truck bed 6.5' x 4.2')
8. GPS and camera integration with acceptable video quality
9. In-flight data must be given via remote terminal

Project Plane



25-26 Design Parameters

Wingspan	15.5 ft 4.7m
Aspect Ratio	18.8
Nose to Tail Length	94in 2.4m
Top Speed (est.)	50mph 22m/s
Lift to Drag Ratio (Cruise)	15
Aircraft Mass	15lb 6.8kg
Max Solar Power Generation	6 watts * 26 panels = 156 watts
Total Cruise Power Draw	200w

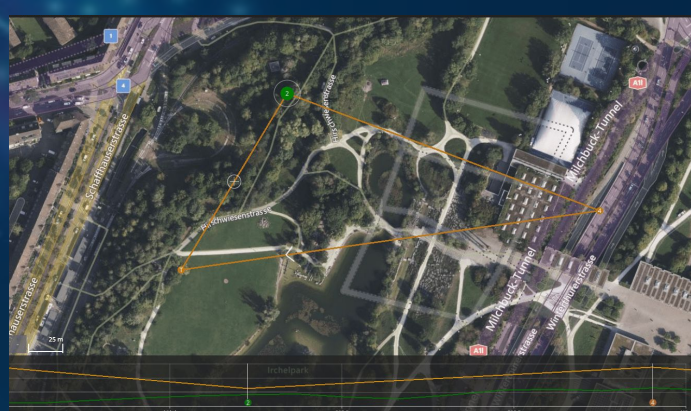
Propulsion

Telemetry Features

Live location & flight data tracking

GPS + Pixhawk

Flight Modes

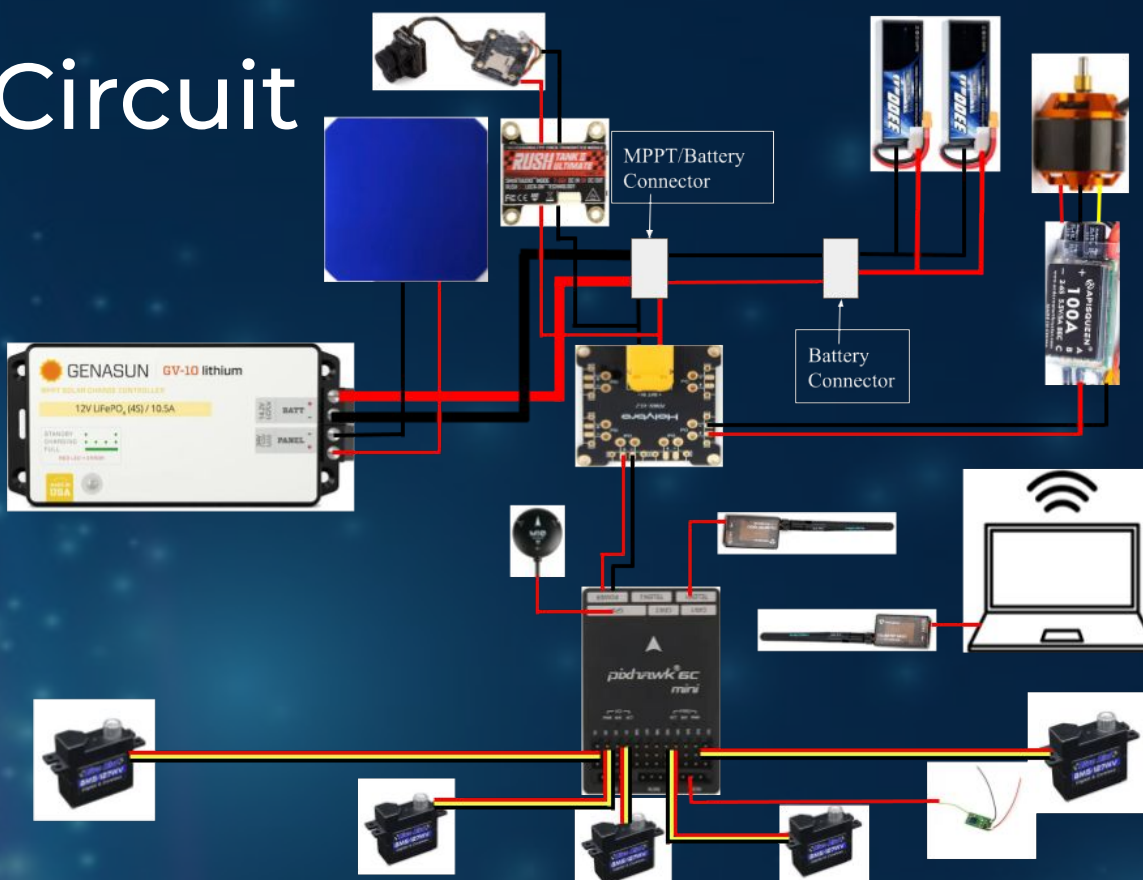


- Manual - Normal RC
- FBWA (Fly By Wire_A) - Assisted Flying
- Loiter - Circles a point and maintains altitude
- Auto - Follows a mission set by ground station
- Guided - Seeks a map point without a mission plan already set
- RTL (Return To Launch) - Returns to home location and loiter

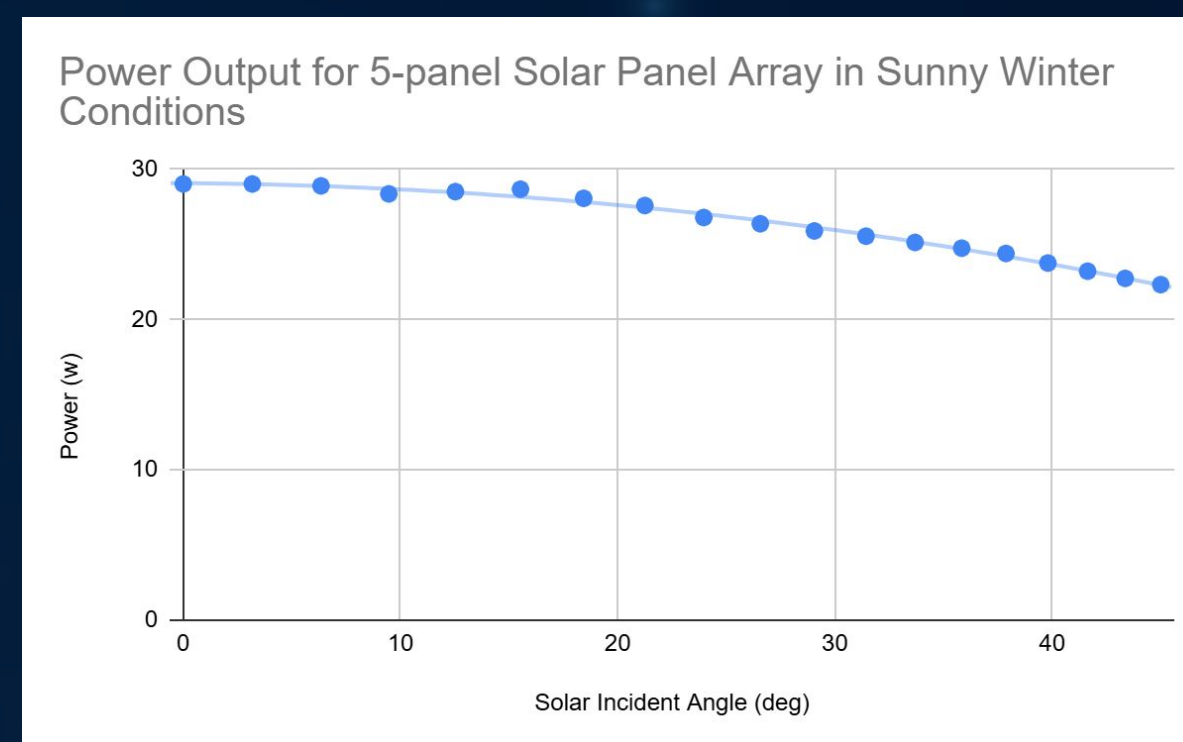
Full Electrical Circuit

Features:

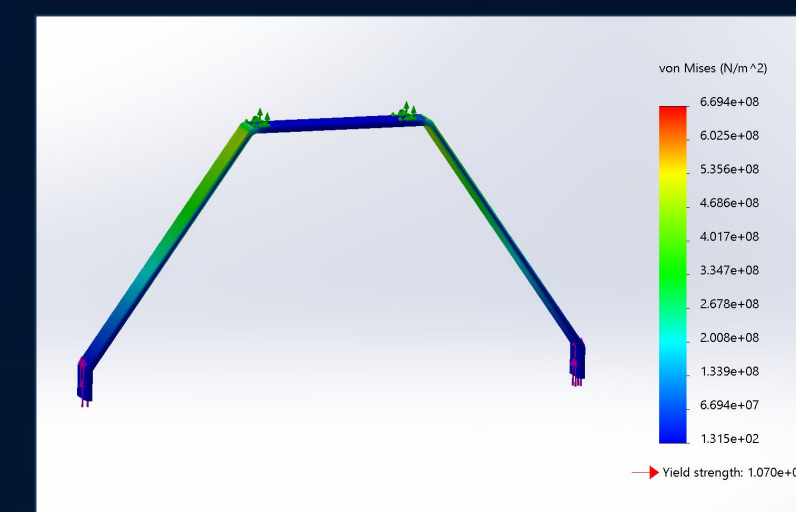
- Solar Power Generation
- Propulsion Systems
- Camera Detection and Live monitoring



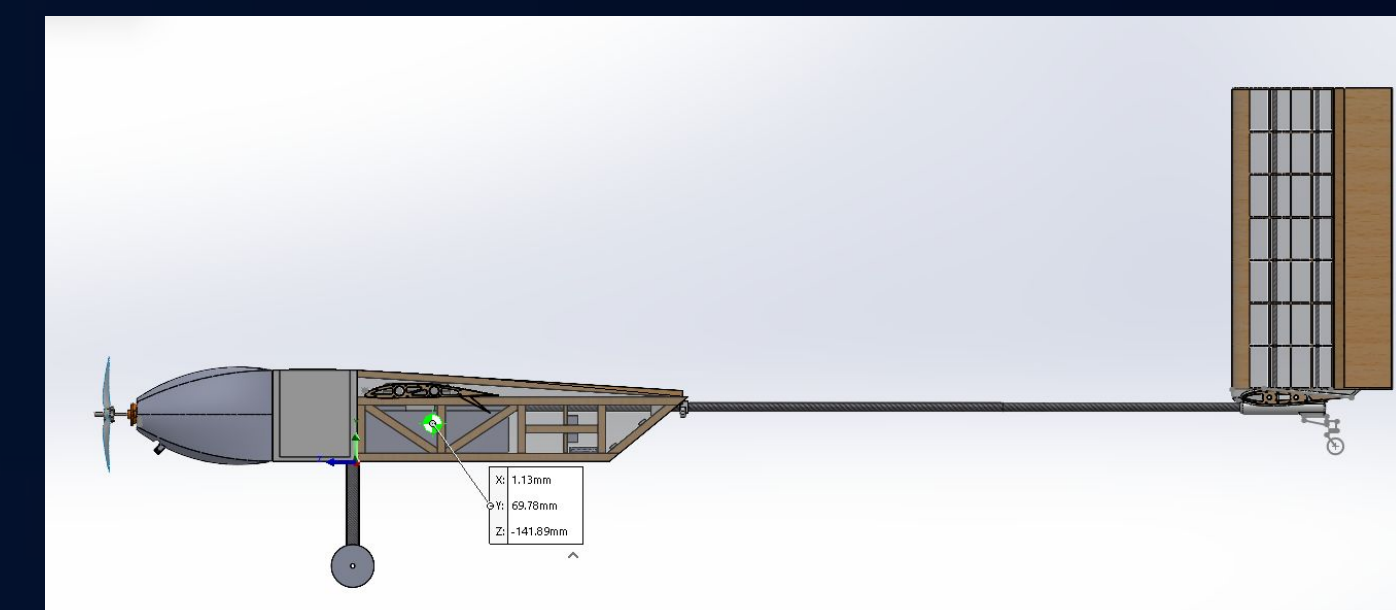
Design and Analysis



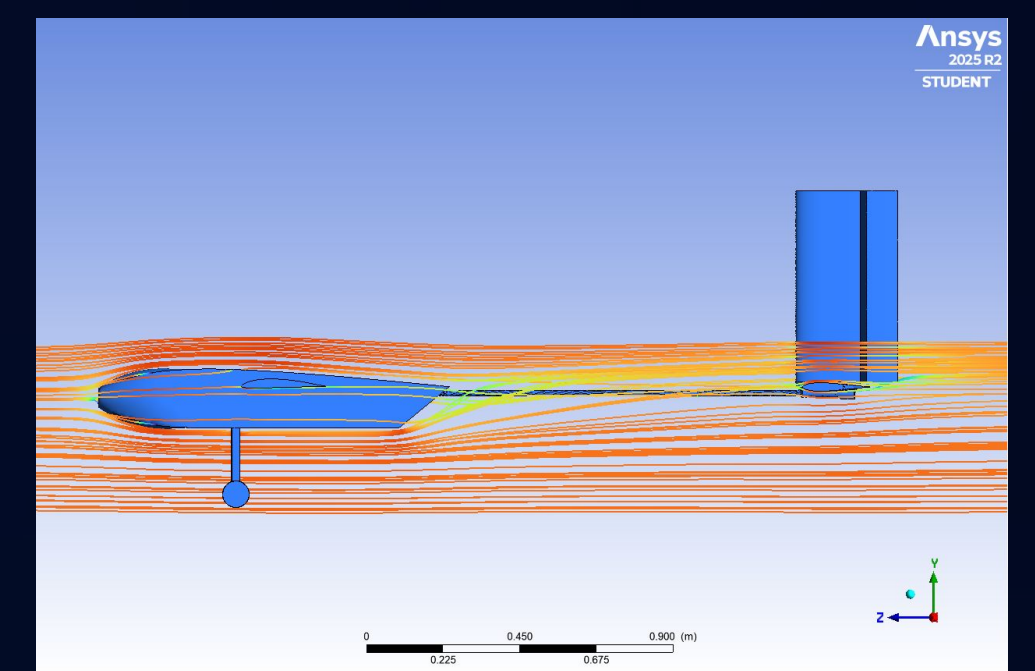
Solar Panel Power Output



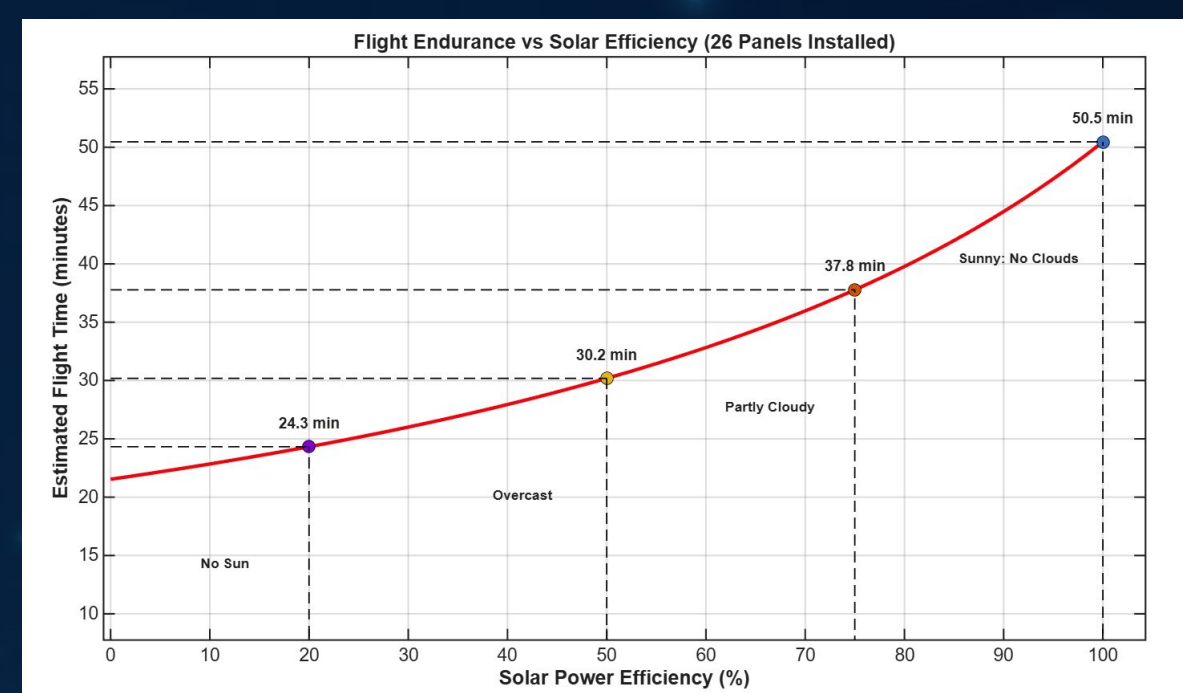
Landing Gear FEA Stress Analysis



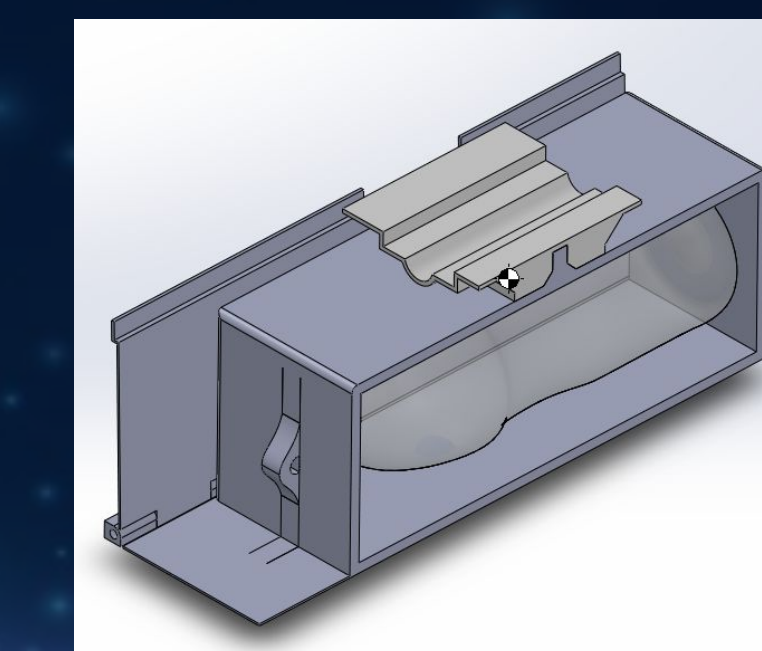
Solidworks CAD Model w/ Center of Mass



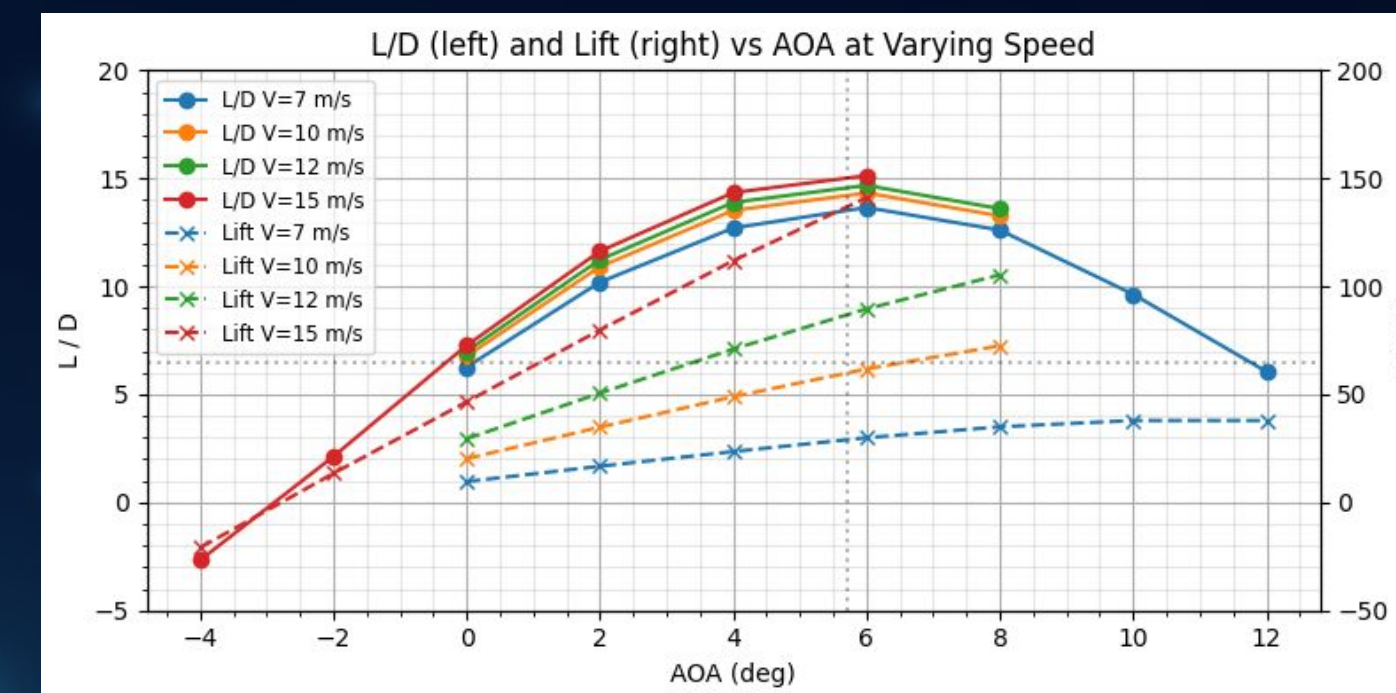
Aircraft CFD - Ansys Fluent



Flight Time Based on Solar Efficiency



Payload Bay mechanism



Flight Efficiency



Tail FEA



Wing Airfoil (S7055)

Team Structure

LinkedIn: "UCI Solar Airplane Project"

Team Leads - Emmanuel Bivian (Project Manager), Patrick Nguyen (Chief Engineer), Graham Richards (Aerodynamics Lead), Brian Chen (Fuselage Lead), Aditya Karki (Propulsion Lead), Michael Simonian (Propulsion Lead)

Subteam Members - **Aerodynamics:** Timothy Yeng, Promise Reid, Joseph Huynh, Aita Sharifi, Taran Joseph Merwin, Jason Lee. **Fuselage:** Peter Huang, Gina Kim, Anthony Fabros, Ting Chen, Sarghi Kamboj, Olivia Wadley. **Propulsion:** Ishika Narain, Brandon Hernandez, Sriharshini Gubbala, Ayush Chanda. **Operations:** Aman Rampure, Vilca Quezada.